

VISHWA G S

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EXECUTIVE SUMMARY

Mechatronics Engineering student with hands-on experience building robotic systems from simulation to physical deployment using ROS2, Nav2, and SLAM Toolbox. Experienced in robot modeling, multi-sensor integration (LiDAR, camera, IMU, encoders), autonomous navigation, and industrial automation using PLCs and IoT technologies. Comfortable developing and testing the full robotics software stack on Ubuntu Linux with Git-based workflows.

TECHNICAL SKILLS

Robotics & Control: ROS2 (Humble), Nav2 Navigation Stack, SLAM Toolbox, TF2, URDF Modeling, Robot Kinematics

Simulation: Gazebo, RViz

Programming: Python, C++

Sensors & Perception: RPLiDAR, IMU, Wheel Encoders, Sensor Fusion, OpenCV, YOLOv8, Camera-to-Robot Coordinate Transformation

Embedded & Automation: ESP32, PLC Programming, OPC UA, MQTT, Node-RED

Tools & Platforms: Git & GitHub, Ubuntu Linux

ROBOTICS PROJECTS

ROS2-Based Autonomous Mobile Robot (AMR)

Feb 2026 – Present

- Designed a differential-drive AMR from the ground up — robot modeling in URDF, kinematics validation in Gazebo, and deployment on physical hardware.
- Built the sensor and control stack: TF transformations, robot state publishing, and wheel encoder-based odometry for accurate pose estimation.
- Achieved real-time occupancy-grid mapping with SLAM Toolbox, upgrading the LiDAR from LDS-01 to RPLiDAR A2 M8 for extended range and accuracy.
- Implemented autonomous navigation using the ROS2 Nav2 stack with obstacle avoidance, path planning, and goal-based navigation, validated on physical hardware.
- Developed a PyQt-based dashboard for real-time robot monitoring, teleoperation, and navigation control.
- Developed and tested the complete robotics software stack on Ubuntu Linux, using Git for version control.

Technologies: ROS2 (Humble), Nav2, SLAM Toolbox, Gazebo, RViz, URDF, RPLiDAR A2 M8, PyQt, Python, Ubuntu Linux, Git

MachineX3 — Industrial Coolant Monitoring System

Jul 2025 – Sep 2025 (approx.)

- Built an industrial IoT monitoring system using ultrasonic and flow sensors on ESP32, with a PLC handling control logic to meet industrial automation standards.
- Set up OPC UA and MQTT communication for reliable sensor-to-dashboard data flow.
- Created a Node-RED dashboard with live alarms/status alerts and automated Google Sheets reporting.

Technologies: ESP32, PLC, OPC UA, MQTT, Node-RED, Google Sheets API

Object Detection & Coordinate Estimation for Robotic Manipulation *Dec 2025 – Jan 2026 (approx. 20 days)*

- Built a camera-based perception pipeline to detect objects and estimate spatial (X, Y, Z) coordinates using computer vision.
- Validated the full camera-to-robot coordinate transformation pipeline, feeding detections into robotic arm motion planning for simulated pick-and-place.

Technologies: Python, OpenCV, YOLOv8, Robot Manipulation Simulation

EDUCATION

Bachelor of Engineering (B.E.) — Mechatronics Engineering

Expected 2028

Bannari Amman Institute of Technology | CGPA: 8.7

CERTIFICATIONS

- NPTEL Certification — Mechatronics

AWARDS & ACCOMPLISHMENTS

- Displayed Autonomous Mobile Robot prototype at CODISSIA Intec.
- Won “Most Innovative Design” at Cadethon.
- Won First Prize at BITHACK Season 1.